Blis Global SARL
Shared Bright Future.

Introduction to Microcontrollers

Student Guide

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Introduction

This document is showing the list of experiments and procedures which will be conducted by students in this course

As Facilitator you should be in advance in the following topics:

Week 1: Introduction to Microcontrollers and Arduino

Topics:

- Understanding Microcontrollers
  - What is a microcontroller?
  - Common applications of microcontrollers
- Introduction to Arduino
  - What is Arduino?
  - Overview of Arduino boards (Uno, Nano, Mega, etc.)
- Setting up the Arduino IDE
  - Installing the Arduino IDE
  - Setting up the Arduino board
  - Uploading the first program (Blink LED)

Hands-On Activity:

- Install the Arduino IDE
- Connect the Arduino board to a computer
- Upload and modify the Blink sketch

Course Title: Introduction to Microcontrollers and Arduino

Objective

By the end of this lesson, students will:

1. Understand the basic concept of microcontrollers.
2. Be familiar with the Arduino platform and its applications.
3. Learn how to set up an Arduino and write a simple program.

Part 1: What is a Microcontroller? (10 minutes)

Student Reading

A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system. Microcontrollers are like small computers that can be programmed to perform tasks. They are used in a wide range of applications, from household appliances like microwaves and washing machines to complex systems like robots and automotive controls.

Microcontrollers contain:

- **CPU (Central Processing Unit)**: The brain of the microcontroller that processes instructions.
- **Memory**: Includes RAM (for temporary data storage) and ROM/Flash memory (for storing programs).
- **Input/Output (I/O) Ports**: Interfaces to connect the microcontroller to other devices, sensors, and actuators.
- **Timers and Counters**: For keeping track of time and events.
- **Communication Interfaces**: Such as UART, SPI, and I2C, which allow the microcontroller to communicate with other devices.

Why are Microcontrollers Important? Microcontrollers are essential for automating tasks in embedded systems. They can execute a series of programmed instructions, allowing machines to perform tasks automatically, respond to sensor inputs, or control other devices.

Part 2: Introduction to Arduino (10 minutes)

Student Reading

Arduino is an open-source electronics platform based on easy-to-use hardware and software. It consists of a programmable circuit board (microcontroller) and a software environment (IDE) for writing and uploading code to the board.

Key Features of Arduino:

- **Microcontroller**: The heart of the Arduino board, usually an AVR-based microcontroller (like ATmega328 on Arduino Uno).
- **Digital and Analog I/O Pins**: Arduino boards have multiple pins for connecting sensors, LEDs, motors, and other components.
- **Power Supply**: Can be powered through USB or an external power supply.
- **Programming Language**: Arduino uses a simplified version of C++.
- **USB Interface**: For connecting to a computer to upload code.

Why Use Arduino? Arduino is popular for prototyping because it's easy to use, even for beginners. It has a large community, which means plenty of tutorials, libraries, and project ideas are available. With Arduino, you can build interactive projects, control motors, read sensor data, and much more.

Part 3: Setting Up Arduino and Writing Your First Program (10 minutes)

Step 1: Setting Up the Arduino IDE

1. **Download and Install**: Go to the Arduino website and download the IDE for your operating system. Install it on your computer.
2. **Connect the Arduino Board**: Use a USB cable to connect the Arduino board (e.g., Arduino Uno) to your computer.
3. **Select the Board and Port**: Open the Arduino IDE. Go to Tools > Board and select your board model (e.g., Arduino Uno). Then, go to Tools > Port and select the port where your Arduino is connected.

Step 2: Writing Your First Program (Blinking an LED)

1. **Open the Example Sketch**: In the Arduino IDE, go to File > Examples > 01.Basics > Blink.
2. **Understanding the Code**:
    - void setup(): This function runs once when you press reset or power the board. It's used to initialize settings.
    - void loop(): This function runs repeatedly in a loop, making your program run continuously.
    - The code in the Blink example turns an LED on and off at one-second intervals.

```cpp
void setup() {
  pinMode(LED_BUILTIN, OUTPUT); // Initialize the LED pin as an output
}

void loop() {
  digitalWrite(LED_BUILTIN, HIGH); // Turn the LED on
  delay(1000);                     // Wait for a second
  digitalWrite(LED_BUILTIN, LOW);  // Turn the LED off
  delay(1000);                     // Wait for a second
}
```

3. **Upload the Code**: Click the Upload button (arrow pointing to the right). The code will be compiled and sent to the Arduino board. The onboard LED should start blinking.
4. **Experiment**: Try changing the delay values to make the LED blink faster or slower.

Conclusion

Microcontrollers like those used in Arduino are powerful tools for automating tasks and building interactive projects. With Arduino, you can quickly prototype and experiment with electronics, making it an ideal platform for beginners and advanced users alike. By understanding the basics of microcontrollers and learning how to use Arduino, you open up a world of possibilities in the field of electronics and robotics.

This lesson provides a basic understanding of microcontrollers and Arduino, giving students the foundation to explore more complex projects in the future.